

The `Square` class starts by inheriting from the `Shape` class ❶. The `__init__()` method calls the `__init__()` method of its base class (or superclass) ❷, identifying this shape as a square and randomly choosing its size.

Next we have three methods whose implementation is specific to a square. The `clickedInside()` method only needs to make a call to `rect.collidepoint()` to determine if a click happened inside its rectangle ❸. The `getArea()` method simply multiplies the `widthAndHeight` by the `widthAndHeight` ❹. Finally, the `draw()` method draws a rectangle using the value of `widthAndHeight` ❺.

[*Listing 10-8*](#) shows the `Circle` class, which has also been modified to inherit from the `Shape` class.

File: InheritedShapes/Circle.py

```
# Circle class

import pygame
from Shape import *
import math

class Circle(Shape):

    def __init__(self, window, maxWidth, maxHeight):
        super().__init__(window, 'Circle', maxWidth, maxHeight)
        self.radius = random.randrange(10, 50)
        self.centerX = self.x + self.radius
        self.centerY = self.y + self.radius
        self.rect = pygame.Rect(self.x, self.y, self.radius * 2, self.radius * 2)

    def clickedInside(self, mousePoint):
        theDistance = math.sqrt(((mousePoint[0] - self.centerX) ** 2) +
                                ((mousePoint[1] - self.centerY) ** 2))
        if theDistance <= self.radius:
            return True
        else:
            return False

    def getArea(self):
        theArea = math.pi * (self.radius ** 2)
        return theArea
```